



Balancing atoms on LHS, RHS

~~$$6 = a + b$$

$$18a + 3c$$~~

$$6 = a + b \rightarrow \text{carbon}$$

$$15 = 18a + 3c \rightarrow \text{H}$$

$$8 = 0.5a + 2b \rightarrow \text{O}$$

$$1 = 0.2a + c \rightarrow \text{N}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 1.8 & 0 & 3 \\ 0.5 & 2 & 0 \\ 0.2 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 15 \\ 8 \\ 1 \end{bmatrix}$$